CHEM 1301 Driller 9-2 — Answer Key

Worked solutions with course-material citations

Every numeric answer was recomputed by `_scripts/build_driller_9_2.py`, which builds this key and the driller from the same data. Atomic masses are periodic-table values (H 1.008, C 12.01, N 14.01, O 16.00, …). Citations name the slide (same number in the `.pdf` and `.pptx`) where the content is taught, plus the closest homework screenshot.

How to read the solutions. Each calculation is a chain of conversion factors written want-over-have. The label under each factor says where its numbers come from (a molar mass, or two coefficients from the balanced equation), and the crossed-out units are the ones that cancel. Study guide section 9.0 explains this format.

Score bands. 14–15 solid on 8.7 and Chapter 9 · 11–13 rework the study guide sections cited for the questions you missed · 10 or below go back through study guide sections 8.7 and 9.2–9.7 before the next driller.

Q	Ans	Answer	Type	Topic (source)
1	**a**	The molecular formula is always a whole-number multiple of the empirical formula	concept	Empirical vs molecular formula (Ch 8 sl. 39–40, 48, 50)
2	**d**	C_3H_7	calc	Molecular to empirical formula (Ch 8 sl. 39–40)
3	**b**	N_2O_5	calc	Percent composition to empirical formula (×2) (Ch 8 sl. 35–38, 42–43)
4	**a**	CH_3O	calc	Percent composition to empirical formula (Ch 8 sl. 35–38, 42)
5	**d**	SO_3	calc	Grams to empirical formula (Ch 8 sl. 45–46)
6	**c**	C_4H_{10}	calc	Empirical to molecular formula (Ch 8 sl. 47–49)
7	**e**	1.50 mol	calc	From percent composition to a mole ratio (Ch 8 sl. 42–49; Ch 9 sl. 9–11)
8	**b**	is completely used up first and so determines how much product can form	concept	Definition of the limiting reactant (Ch 9 sl. 21–22)
9	**c**	2.25 mol	calc	Mole–mole conversion (Ch 9 sl. 10–11)

Q	Ans	Answer	Type	Topic (source)
10	**e**	45.0 g	calc	Moles to grams of product (Ch 9 sl. 12, 14, 17)
11	**b**	0.308 g	calc	Grams to grams (Ch 9 sl. 14, 18–20)
12	**e**	35.3 g	calc	Balance first, then grams to grams (Ch 9 sl. 14, 16, 19)
13	**a**	O_2 is limiting, and 1.98 g of CH_4 is left over	calc	Limiting reactant and leftover (Ch 9 sl. 21–24)
14	**c**	23.0 g	calc	Theoretical yield from two masses (Ch 9 sl. 24–27)
15	**d**	92.9%	calc	Percent yield (Ch 9 sl. 28–32)

Mix: 2 concept / 13 calculation · Ch 8: 6 · Ch 9: 9

1. Empirical vs molecular formula — a) The molecular formula is always a whole-number multiple of the empirical formula

Source: Chapter 8 post, slide 39 (empirical vs molecular formulas), slide 48 (molecular = $(\text{empirical})_n$), slide 50
Homework match: HW21 — hw21ex3.png (CH with molar mass 26 gives C_2H_2)
What it tests: Course definitions: empirical = smallest whole-number ratio; molecular = $(\text{empirical})_n$, n a whole number.

Definitions (study guide 8.7). The empirical formula is the **smallest whole-number ratio** of atoms. The molecular formula is the **actual** number of atoms in one molecule. They are linked by **molecular = $(\text{empirical})_n$**, where n is a whole number (1, 2, 3, …), as slide 48 states.

So the molecular formula is always the empirical formula times a whole number. When n = 1 the two formulas are the same (H_2O, CO_2).

Why the other choices are wrong:

- b) The empirical formula gives the actual number of atoms in one molecule: That is the **molecular** formula (slide 50). The empirical formula only gives the smallest ratio.
- c) The empirical formula is always different from the molecular formula: They can be the same. For H_2O, n = 1, so the empirical and molecular formulas are both H_2O.
- d) An empirical formula may contain fractional subscripts such as $CH_{1.5}$: Empirical formulas use the smallest **whole-number** ratio. A .5 means multiply every subscript by 2.
- e) Two different compounds can never have the same empirical formula: They can: C_2H_2 and C_6H_6 are both CH (slide 39).

2. Molecular to empirical formula — d) C_3H_7

Source: Chapter 8 post, slide 39 (empirical formula = smallest whole-number ratio), slide 40
Homework match: HW21 — hw21ex3.png (relating empirical and molecular formulas)
What it tests: Type A (study guide 8.7): divide every subscript by their largest common divisor.

Step 1: list the subscripts. C 6, H 14.

Step 2: find the largest number that divides both evenly. $6 \div 2 = 3$ and $14 \div 2 = 7$. No larger number divides both (3 doesn't divide 14, and 7 doesn't divide 6), so the divisor is **2**.

Step 3: divide every subscript by 2. C $6 \div 2 = 3$, H $14 \div 2 = 7$. Empirical formula: $\mathbf{C_3H_7}$.

Why the other choices are wrong:

- a) CH_2: That reduces C_6H_{12}, not C_6H_{14}. 14 isn't divisible by 6.
- b) C_2H_5: Divided C by 3, but $14 \div 3$ isn't a whole number. Every subscript must be divided by the **same** number.
- c) C_6H_{14}: That is the molecular formula. 6 and 14 are both divisible by 2, so it isn't the smallest ratio.
- e) C_3H_8: C was divided by 2, but $14 \div 2 = 7$, not 8.

3. Percent composition to empirical formula (×2) — b) N_2O_5

Source: Chapter 8 post, slides 42–43 (Steps for Determining the Empirical Formula; step 4: multiply when not whole), slides 35–38
Homework match: HW21 — hw21ex2.png (empirical formula from percent composition: $MgSO_4$, $SiCl_2Br_2$)
What it tests: The 4-step method: grams → moles → divide by the smallest → whole numbers (multiply every element if needed).

Step 1: percent to grams. Pretend you have exactly **100 g** of the compound, so each percent becomes that many grams: 25.94 g N, 74.06 g O.

Step 2: grams to moles for each element. Divide by each element's own atomic mass (periodic table):

$$25.94 \cancel{\text{ g N}} \times \underbrace{\frac{1 \text{ mol N}}{14.01 \cancel{\text{ g N}}}}_{\text{atomic mass of N}} = 1.852 \text{ mol N}$$

$$74.06 \cancel{\text{ g O}} \times \underbrace{\frac{1 \text{ mol O}}{16.00 \cancel{\text{ g O}}}}_{\text{atomic mass of O}} = 4.629 \text{ mol O}$$

Step 3: divide every mole value by the smallest one. The smallest mole value is 1.852 mol (N). Divide **each** element's moles by it:

Element	Moles (step 2)	÷ smallest (1.852 mol N)	× 2	Subscript
N	1.852	1.00	2.00	2
O	4.629	2.50	5.00	5

Step 4: whole numbers? 2.50 is not within 0.1 of a whole number. It ends in a decimal near .5 (halves), so multiply **every** result by 2, including the elements that were already whole (last columns of the table). Empirical formula: $\mathbf{N_2O_5}$.

Why the other choices are wrong:

- a) NO_5: Multiplied only the fractional element by 2. The same multiplier goes on **every** element.
- c) NO_3: Used the masses (percents) directly instead of converting to moles first.
- d) N_4O_{10}: Not the **smallest** whole-number ratio; every subscript can still be divided by 2.
- e) NO_2: Rounded the fractional result instead of multiplying (only results within 0.1 of a whole number may be rounded).

4. Percent composition to empirical formula — a) CH_3O

Source: Chapter 8 post, slides 35–37 (four-element example: percent → grams → moles → ratio), slide 42
Homework match: HW21 — hw21ex2.png (empirical formula from percent composition)
What it tests: The 4-step method: grams → moles → divide by the smallest → whole numbers (multiply every element if needed).

Step 1: percent to grams. Pretend you have exactly **100 g** of the compound, so each percent becomes that many grams: 38.70 g C, 9.74 g H, 51.56 g O.

Step 2: grams to moles for each element. Divide by each element's own atomic mass (periodic table):

$$38.70 \cancel{\text{g C}} \times \underbrace{\frac{1 \text{ mol C}}{12.01 \cancel{\text{g C}}}}_{\text{atomic mass of C}} = 3.222 \text{ mol C}$$

$$9.74 \cancel{\text{g H}} \times \underbrace{\frac{1 \text{ mol H}}{1.008 \cancel{\text{g H}}}}_{\text{atomic mass of H}} = 9.663 \text{ mol H}$$

$$51.56 \cancel{\text{g O}} \times \underbrace{\frac{1 \text{ mol O}}{16.00 \cancel{\text{g O}}}}_{\text{atomic mass of O}} = 3.223 \text{ mol O}$$

Step 3: divide every mole value by the smallest one. The smallest mole value is 3.222 mol (C). Divide **each** element's moles by it:

Element	Moles (step 2)	÷ smallest (3.222 mol C)	Subscript
C	3.222	1.00	1
H	9.663	3.00	3
O	3.223	1.00	1

Step 4: whole numbers? Every result is within 0.1 of a whole number, so round them. No multiplier is needed. Empirical formula: $\mathbf{CH_3O}$.

Why the other choices are wrong:

- b) CH_4O: Miscounted H: its step-3 result is 3.00, which gives 3, not 4.
- c) C_4HO_5: Used the masses (percents) directly instead of converting to moles first.
- d) C_2H_3O: Miscounted C: its step-3 result is 1.00, which gives 1, not 2.
- e) $C_2H_6O_2$: Not the **smallest** whole-number ratio; every subscript can still be divided by 2.

5. Grams to empirical formula — d) SO_3

Source: Chapter 8 post, slides 45–46 (empirical formula from masses: 1.198 g C + 0.302 g H), slide 42
Homework match: HW21 — empirical formula from composition data
What it tests: The 4-step method: grams → moles → divide by the smallest → whole numbers (multiply every element if needed).

Step 1: grams of each element. Sulfur: 2.00 g (given). Oxygen is the rest of the sample: 5.00 g − 2.00 g = **3.00 g O**.

Step 2: grams to moles for each element. Divide by each element's own atomic mass (periodic table):

$$2.00 \cancel{\text{g S}} \times \underbrace{\frac{1 \text{ mol S}}{32.07 \cancel{\text{g S}}}}_{\text{atomic mass of S}} = 0.06236 \text{ mol S}$$

$$3.00 \cancel{\text{g O}} \times \underbrace{\frac{1 \text{ mol O}}{16.00 \cancel{\text{g O}}}}_{\text{atomic mass of O}} = 0.1875 \text{ mol O}$$

Step 3: divide every mole value by the smallest one. The smallest mole value is 0.06236 mol (S). Divide **each** element's moles by it:

Element	Moles (step 2)	÷ smallest (0.06236 mol S)	Subscript
S	0.06236	1.00	1
O	0.1875	3.01	3

Step 4: whole numbers? Every result is within 0.1 of a whole number, so round them. No multiplier is needed. Empirical formula: **SO_3**.

Why the other choices are wrong:

- a) S_2O_6: Not the **smallest** whole-number ratio; every subscript can still be divided by 2.
- b) SO_2: Used the masses (percents) directly instead of converting to moles first.
- c) S_3O: Subscripts swapped between the elements.
- e) S_2O_3: Miscounted S: its step-3 result is 1.00, which gives 1, not 2.

6. Empirical to molecular formula — c) C_4H_{10}

Source: Chapter 8 post, slide 48 (molecular formula = (empirical formula)$_n$), slide 49 (LA Practice), slide 47
Homework match: HW21 — hw21ex3.png (CH, molar mass 26 → C_2H_2; C_3H_7NO → $C_6H_{14}N_2O_2$)
What it tests: Molecular formula = (empirical formula)$_n$ with n = molar mass ÷ empirical formula mass.

Step 1: empirical formula mass. Add up the atomic masses in the empirical formula: C_2H_5 = 2(12.01) + 5(1.008) = 29.06 g/mol.

Step 2: find n.

$$n = \frac{\text{molar mass of the compound}}{\text{empirical formula mass}} = \frac{58.12 \text{ g/mol}}{29.06 \text{ g/mol}} = 2.00 \approx 2$$

Step 3: multiply every subscript by n. $(C_2H_5)_2$: C $2 \times 2 = 4$, H $5 \times 2 = 10$. Molecular formula: $\mathbf{C_4H_{10}}$.

Why the other choices are wrong:

- a) C_4H_5: Multiplied only the first subscript by n = 2. **Every** subscript gets multiplied.
- b) C_8H_{20}: n = 4 would give 116.24 g/mol.
- d) C_6H_{15}: n = 3 would give a molar mass of 87.18 g/mol.
- e) C_2H_5: That is the empirical formula itself (n = 1). Its molar mass would be 29.06 g/mol, not 58.12.

7. From percent composition to a mole ratio — e) 1.50 mol

Source: Chapter 8 post, slides 42–49 (empirical then molecular formula); Chapter 9 post, slide 9 (balancing a combustion), slides 10–11
Homework match: HW21 — hw21ex3.png (empirical + molar mass → molecular); HW22 — hw22ex3.png (moles of one substance from another)
What it tests: Section 8.7 feeding Chapter 9: percent composition → empirical → molecular formula → balanced equation → mole ratio.

Step 1: percent to grams. Pretend you have exactly **100 g** of the compound, so each percent becomes that many grams: 85.63 g C, 14.37 g H.

Step 2: grams to moles for each element. Divide by each element's own atomic mass (periodic table):

$$85.63 \cancel{\text{g C}} \times \underbrace{\frac{1 \text{ mol C}}{12.01 \cancel{\text{g C}}}}_{\text{atomic mass of C}} = 7.130 \text{ mol C}$$

$$14.37 \cancel{\text{g H}} \times \underbrace{\frac{1 \text{ mol H}}{1.008 \cancel{\text{g H}}}}_{\text{atomic mass of H}} = 14.26 \text{ mol H}$$

Step 3: divide every mole value by the smallest one. The smallest mole value is 7.130 mol (C). Divide **each** element's moles by it:

Element	Moles (step 2)	÷ smallest (7.130 mol C)	Subscript
C	7.130	1.00	1
H	14.26	2.00	2

Step 4: whole numbers? Every result is within 0.1 of a whole number, so round them. No multiplier is needed. Empirical formula: $\mathbf{CH_2}$.

Step 5: molecular formula. Empirical formula mass: $CH_2 = 12.01 + 2(1.008) = 14.03$ g/mol. n = 28.05 ÷ 14.03 = 2.00 ≈ 2, so multiply every subscript by 2: $\mathbf{C_2H_4}$.

Step 6: write the balanced equation. Combustion of C_2H_4: balance C first ($2CO_2$), then H ($2H_2O$, 4 H), then O: the right side has 2(2) + 2(1) = 6 O, so $3O_2$. $\mathbf{C_2H_4 + 3O_2 \rightarrow 2CO_2 + 2H_2O}$.

Step 7: moles $\mathbf{C_2H_4}$ to moles $\mathbf{O_2}$. Have C_2H_4 (coefficient **1**), want O_2 (coefficient **3**).

$$0.5000 \text{ mol } C_2H_4 \times \underbrace{\frac{3 \text{ mol } O_2}{1 \text{ mol } C_2H_4}}_{\text{coefficient of } O_2 \text{ / coefficient of } C_2H_4} = \boxed{1.50 \text{ mol } O_2}$$

The molecular formula matters here. A mole of the **empirical** unit is a different amount of substance from a mole of the real molecule, so it would give a different number of moles of product.

Why the other choices are wrong:

- a) 0.167 mol: Ratio upside down: × 1/3 instead of × 3/1.
- b) 0.500 mol: Assumed 1 : 1. O_2's coefficient is 3.
- c) 0.750 mol: Used the empirical formula CH_2 ($CH_2 + 1.5O_2$), which is half a molecule.
- d) 1.00 mol: Used CO_2's coefficient (2) instead of O_2's (3).

8. Definition of the limiting reactant — b) is completely used up first and so determines how much product can form

Source: Chapter 9 post, slide 21 (Limiting Reactants: the reactant that runs out first), slide 22, slide 23
Homework match: HW23 — hw23ex2.png (determine which reactant is limiting)
What it tests: Course definition of the limiting reactant, and why mass, coefficients, or raw moles can't identify it.

Slide 21: the limiting reactant is the reactant that **runs out first**, and it **limits the amount of product** that can form. The other reactant is in excess and some of it is left over.

To find it, compare what one reactant **needs** of the other (through the mole ratio) with what is actually **available** (study guide 9.5).

Why the other choices are wrong:

- a) is left over when the reaction stops: That is the **excess** reactant.
- c) has the fewest moles present: Raw moles don't decide it either. 3Mg + N_2 on slide 23 has more moles of Mg, yet Mg is limiting.
- d) has the smallest coefficient in the balanced equation: Coefficients alone don't decide it. You need the actual moles present compared through the ratio.
- e) has the smallest mass: Mass alone doesn't decide it. A reactant with a smaller mass can be in excess.

9. Mole–mole conversion — c) 2.25 mol

Source: Chapter 9 post, slides 10–11 (Mole–Mole Conversions Practice)
Homework match: HW22 — hw22ex3.png (moles of product from moles of another substance)
What it tests: One step: moles given × (coefficient wanted / coefficient given).

Step 1: moles $\mathbf{NaN_3}$ to moles $\mathbf{N_2}$. Have NaN_3 (coefficient **2**), want N_2 (coefficient **3**). The question asks for **moles**, so stop here.

$$1.500 \cancel{\text{ mol NaN}_3} \times \underbrace{\frac{3 \text{ mol N}_2}{2 \cancel{\text{ mol NaN}_3}}}_{\text{coefficient of N}_2 \text{ / coefficient of NaN}_3} = \boxed{2.25 \text{ mol N}_2}$$

Why the other choices are wrong:

- a) 1.00 mol: Mole ratio upside down: $\times$ 2/3 instead of $\times$ 3/2.
- b) 1.50 mol: Assumed 1 : 1. The coefficients are 2 (NaN_3) and 3 (N_2).
- d) 9.00 mol: Multiplied by both coefficients ($\times$ 3 $\times$ 2) instead of $\times$ 3/2.
- e) 63.0 mol: Multiplied by the molar mass of N_2. That gives grams, not moles.

10. Moles to grams of product — e) 45.0 g

Source: Chapter 9 post, slide 17 (Stoichiometry Practice 1: grams of SO_3 from 2.33 mol O_2), slide 8 (N_2 + $O_2 \rightarrow$ 2NO), slide 14
Homework match: HW23 — mass of product from an amount of reactant
What it tests: mol $\rightarrow$ mol $\rightarrow$ mole ratio $\rightarrow$ g.

Step 1: molar masses. NO = 14.01 + 16.00 = 30.01 g/mol.

Step 2: moles N_2 to moles NO. Have N_2 (coefficient **1**), want NO (coefficient **2**).

$$0.7500 \cancel{\text{ mol N}_2} \times \underbrace{\frac{2 \text{ mol NO}}{1 \cancel{\text{ mol N}_2}}}_{\text{coefficient of NO / coefficient of N}_2} = 1.500 \text{ mol NO}$$

Step 3: moles NO to grams NO.

$$1.500 \cancel{\text{ mol NO}} \times \underbrace{\frac{30.01 \text{ g NO}}{1 \cancel{\text{ mol NO}}}}_{\text{molar mass of NO}} = \boxed{45.0 \text{ g NO}}$$

Why the other choices are wrong:

- a) 1.50 g: That is the **moles** of NO. The last step ($\times$ molar mass) is missing.
- b) 11.3 g: Mole ratio upside down: $\times$ 1/2 instead of $\times$ 2/1.
- c) 22.5 g: Assumed 1 : 1. The coefficients are 1 (N_2) and 2 (NO).
- d) 42.0 g: Used the molar mass of N_2 in the last step instead of NO's.

11. Grams to grams — b) 0.308 g

Source: Chapter 9 post, slides 19–20 (Stoichiometry Practice: g $\rightarrow$ mol $\rightarrow$ mol $\rightarrow$ g), slide 14 (steps)
Homework match: HW23 — hw23ex1.png (grams of C_2H_2 from 3.25 g CaC_2)
What it tests: g $\rightarrow$ mol $\rightarrow$ mole ratio $\rightarrow$ g.

Step 1: molar masses. Zn = 65.38 g/mol (periodic table); H_2 = 2(1.008) = 2.02 g/mol.

Step 2: grams Zn to moles Zn.

$$10.0 \text{ g } \cancel{\text{Zn}} \times \underbrace{\frac{1 \text{ mol Zn}}{65.38 \text{ g } \cancel{\text{Zn}}}}_{\text{molar mass of Zn}} = 0.1530 \text{ mol Zn}$$

Step 3: moles Zn to moles H_2. Have Zn (coefficient **1**), want H_2 (coefficient **1**).

$$0.1530 \text{ } \cancel{\text{mol Zn}} \times \underbrace{\frac{1 \text{ mol } H_2}{1 \text{ } \cancel{\text{mol Zn}}}}_{\text{coefficient of } H_2 \text{ / coefficient of Zn}} = 0.1530 \text{ mol } H_2$$

Step 4: moles H_2 to grams H_2.

$$0.1530 \text{ } \cancel{\text{mol } H_2} \times \underbrace{\frac{2.02 \text{ g } H_2}{1 \text{ } \cancel{\text{mol } H_2}}}_{\text{molar mass of } H_2} = \boxed{0.308 \text{ g } H_2}$$

Why the other choices are wrong:

- a) 0.153 g: That is the **moles** of H_2. The last step (× molar mass) is missing.
- c) 0.617 g: Used HCl's coefficient (2) instead of H_2's (1).
- d) 10.0 g: Used the molar mass of Zn in the last step instead of H_2's.
- e) 13.2 g: Divided the molar mass by the grams (65.38 ÷ 10.0).

12. Balance first, then grams to grams — e) 35.3 g

Source: Chapter 9 post, slide 16 (why the equation must be balanced first), slide 14, slide 19
Homework match: HW22 — hw22ex3.png (unbalanced equations: balance, then convert)
What it tests: g → mol → mole ratio → g, after balancing.

Step 1: balance the equation. $SiCl_4$ has 4 Cl and Cl_2 supplies Cl in pairs, so use **2**Cl_2 (4 Cl). Si is already 1 = 1: **Si + 2Cl_2 → $SiCl_4$**.

Step 2: molar masses. Si = 28.09 g/mol (periodic table); Cl_2 = 2(35.45) = 70.90 g/mol.

Step 3: grams Si to moles Si.

$$7.00 \text{ } \cancel{\text{g Si}} \times \underbrace{\frac{1 \text{ mol Si}}{28.09 \text{ } \cancel{\text{g Si}}}}_{\text{molar mass of Si}} = 0.2492 \text{ mol Si}$$

Step 4: moles Si to moles Cl_2. Have Si (coefficient **1**), want Cl_2 (coefficient **2**).

$$0.2492 \text{ } \cancel{\text{mol Si}} \times \underbrace{\frac{2 \text{ mol } Cl_2}{1 \text{ } \cancel{\text{mol Si}}}}_{\text{coefficient of } Cl_2 \text{ / coefficient of Si}} = 0.4984 \text{ mol } Cl_2$$

Step 5: moles Cl_2 to grams Cl_2.

$$0.4984 \text{ } \cancel{\text{mol } Cl_2} \times \underbrace{\frac{70.90 \text{ g } Cl_2}{1 \text{ } \cancel{\text{mol } Cl_2}}}_{\text{molar mass of } Cl_2} = \boxed{35.3 \text{ g } Cl_2}$$

Why the other choices are wrong:

- a) 0.498 g: That is the **moles** of Cl_2. The last step (× molar mass) is missing.
- b) 8.83 g: Mole ratio upside down: × 1/2 instead of × 2/1.
- c) 14.0 g: Used the molar mass of Si in the last step instead of Cl_2's.
- d) 17.7 g: Assumed 1 : 1. The coefficients are 1 (Si) and 2 (Cl_2).

13. Limiting reactant and leftover — a) O_2 is limiting, and 1.98 g of CH_4 is left over

Source: Chapter 9 post, slide 24 (limiting reactant from two masses), slides 21–23
Homework match: HW23 — hw23ex2.png (5.65 g of each reactant: which is limiting?)
What it tests: Limiting reactant by need-vs-have in moles, then the leftover mass of the excess reactant.

Step 1: molar masses. $CH_4 = 12.01 + 4(1.008) = 16.04$ g/mol; $O_2 = 2(16.00) = 32.00$ g/mol.

Step 2: grams to moles for each reactant.

$$8.00 \text{ g } \cancel{\text{CH}_4} \times \underbrace{\frac{1 \text{ mol CH}_4}{16.04 \text{ g } \cancel{\text{CH}_4}}}_{\text{molar mass of CH}_4} = 0.4987 \text{ mol CH}_4$$

$$24.0 \text{ g } \cancel{\text{O}_2} \times \underbrace{\frac{1 \text{ mol O}_2}{32.00 \text{ g } \cancel{\text{O}_2}}}_{\text{molar mass of O}_2} = 0.7500 \text{ mol O}_2$$

Step 3: how much O_2 does all the CH_4 need? Have CH_4 (coefficient **1**), want O_2 (coefficient **2**).

$$0.4987 \text{ mol } \cancel{\text{CH}_4} \times \underbrace{\frac{2 \text{ mol O}_2}{1 \text{ mol } \cancel{\text{CH}_4}}}_{\text{coefficient of O}_2 \text{ / coefficient of CH}_4} = 0.9974 \text{ mol O}_2$$

Step 4: compare. You **need** 0.9974 mol O_2 but only **have** 0.7500 mol, so **O_2 is limiting** and CH_4 is in excess.

Step 5: how much CH_4 reacts? All 0.7500 mol of the limiting O_2 reacts. Have O_2 (coefficient **2**), want CH_4 (coefficient **1**).

$$0.7500 \text{ mol } \cancel{\text{O}_2} \times \underbrace{\frac{1 \text{ mol CH}_4}{2 \text{ mol } \cancel{\text{O}_2}}}_{\text{coefficient of CH}_4 \text{ / coefficient of O}_2} = 0.3750 \text{ mol CH}_4$$

$$0.3750 \text{ mol } \cancel{\text{CH}_4} \times \underbrace{\frac{16.04 \text{ g CH}_4}{1 \text{ mol } \cancel{\text{CH}_4}}}_{\text{molar mass of CH}_4} = 6.02 \text{ g CH}_4$$

Step 6: left over. 8.00 g CH_4 at the start − 6.02 g that reacts = **1.98 g left over**.

Mass alone never decides the limiting reactant: compare in moles, through the mole ratio.

Why the other choices are wrong:

- b) O_2 is limiting, and 6.02 g of CH_4 is left over: That is the mass of CH_4 that **reacts**. Left over = 8.00 g − 6.02 g.
- c) Neither is limiting; both reactants are used up completely: That needs the moles to be in exactly a 1 : 2 ratio, and they aren't.
- d) O_2 is limiting, and 8.00 g of CH_4 is left over: That is **all** of the CH_4. Some of it reacts with the O_2, so less is left.
- e) CH_4 is limiting, and O_2 is left over: Check what all of one reactant needs. The need-vs-have comparison shows the O_2 runs out first.

14. Theoretical yield from two masses — c) 23.0 g

Source: Chapter 9 post, slide 27 (theoretical yield when both masses are given), slide 25 (definition), slide 24
Homework match: HW23 — hw23ex3.png (find the limiting reactant, then the mass of product); HW24 — hw24ex2.png
What it tests: Moles of each reactant → limiting reactant → product from the limiting reactant only.

Step 1: molar masses. NO = 14.01 + 16.00 = 30.01 g/mol; O_2 = 2(16.00) = 32.00 g/mol; NO_2 = 14.01 + 2(16.00) = 46.01 g/mol.

Step 2: grams to moles for each reactant.

$$15.0 \text{ g NO} \times \underbrace{\frac{1 \text{ mol NO}}{30.01 \text{ g NO}}}_{\text{molar mass of NO}} = 0.4998 \text{ mol NO}$$

$$10.0 \text{ g } O_2 \times \underbrace{\frac{1 \text{ mol } O_2}{32.00 \text{ g } O_2}}_{\text{molar mass of } O_2} = 0.3125 \text{ mol } O_2$$

Step 3: how much O_2 does all the NO need? Have NO (coefficient **2**), want O_2 (coefficient **1**).

$$0.4998 \text{ mol NO} \times \underbrace{\frac{1 \text{ mol } O_2}{2 \text{ mol NO}}}_{\text{coefficient of } O_2 \text{ / coefficient of NO}} = 0.2499 \text{ mol } O_2 \text{ needed}$$

Step 4: compare. You **have** 0.3125 mol O_2 and only **need** 0.2499 mol. O_2 is in excess, so **NO is limiting**.

Step 5: theoretical yield from the limiting reactant (NO). Have NO (coefficient **2**), want NO_2 (coefficient **2**).

$$0.4998 \text{ mol NO} \times \underbrace{\frac{2 \text{ mol } NO_2}{2 \text{ mol NO}}}_{\text{coefficient of } NO_2 \text{ / coefficient of NO}} = 0.4998 \text{ mol } NO_2$$

$$0.4998 \text{ mol } NO_2 \times \underbrace{\frac{46.01 \text{ g } NO_2}{1 \text{ mol } NO_2}}_{\text{molar mass of } NO_2} = \boxed{23.0 \text{ g } NO_2}$$

Cross-check: all the O_2 would make 28.8 g NO_2. The smaller amount, 23.0 g, is the theoretical yield.

Why the other choices are wrong:

- a) 0.500 g: That is the **moles** of NO_2.
- b) 15.0 g: Used the molar mass of NO instead of NO_2 in the last step.
- d) 25.0 g: Added the reactant masses. Part of the excess reactant is left over.
- e) 28.8 g: Calculated from O_2. O_2 is in excess, so this is more than the NO can make.

15. Percent yield — d) 92.9%

Source: Chapter 9 post, slide 30 (percent yield formula), slide 32 ($2KClO_3$ decomposition), slide 28 (actual vs theoretical)
Homework match: HW24 — hw24ex1.png (percent yield from theoretical and actual masses)
What it tests: Two parts: theoretical yield from the limiting reactant, then actual / theoretical × 100.

Step 1: which reactant controls the yield? $NaHCO_3$ is the only reactant, so it is limiting.

Step 2: molar masses. $NaHCO_3 = 22.99 + 1.008 + 12.01 + 3(16.00) = 84.01$ g/mol; $Na_2CO_3 = 2(22.99) + 12.01 + 3(16.00) = 105.99$ g/mol.

Step 3: theoretical yield of Na_2CO_3. Have $NaHCO_3$ (coefficient **2**), want Na_2CO_3 (coefficient **1**).

$$16.8 \text{ g } \cancel{NaHCO_3} \times \underbrace{\frac{1 \text{ mol } NaHCO_3}{84.01 \text{ g } \cancel{NaHCO_3}}}_{\text{molar mass of } NaHCO_3} = 0.2000 \text{ mol } NaHCO_3$$

$$0.2000 \text{ } \cancel{\text{mol } NaHCO_3} \times \underbrace{\frac{1 \text{ mol } Na_2CO_3}{2 \text{ } \cancel{\text{mol } NaHCO_3}}}_{\text{coefficient of } Na_2CO_3 \text{ / coefficient of } NaHCO_3} = 0.09999 \text{ mol } Na_2CO_3$$

$$0.09999 \text{ } \cancel{\text{mol } Na_2CO_3} \times \underbrace{\frac{105.99 \text{ g } Na_2CO_3}{1 \text{ } \cancel{\text{mol } Na_2CO_3}}}_{\text{molar mass of } Na_2CO_3} = 10.60 \text{ g } Na_2CO_3 \text{ (theoretical)}$$

Step 4: percent yield. The actual yield (9.85 g, measured) goes on top, the theoretical yield on the bottom:

$$\text{percent yield} = \frac{9.85 \text{ } \cancel{\text{g}} \text{ (actual)}}{10.60 \text{ } \cancel{\text{g}} \text{ (theoretical)}} \times 100\% = \boxed{92.9\%}$$

Why the other choices are wrong:

- a) 0.929%: Forgot to multiply by 100.
- b) 46.5%: Theoretical yield found with a 1 : 1 ratio. The coefficients are 2 ($NaHCO_3$) and 1 (Na_2CO_3).
- c) 58.6%: Divided by the mass of $NaHCO_3$. The bottom must be the theoretical yield of Na_2CO_3.
- e) 108%: Upside down: theoretical / actual. A yield can't be over 100%.